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46 Data from the American Academy of Sleep Medicine: (bl). **50 PNAS:** Based on Fig. 1 from "A snapshot of the age distribution of psychological well-being in the United States," Arthur A. Stone et al., Proceedings of the National Academy of Sciences Jun 2010, 107 (22) 9985–9990; DOI: 10.1073/

pnas.1003744107 (bl). **51 APA:** (Excluding explanatory annotation): Based on Fig. 2—Longitudinal estimates of age changes in factor scores on six primary mental abilities at the latent construct level. From "The Course of Adult Intellectual Development" by K. W. Schaie 1994, American Psychologist, 49, pp. 304–313 © 1994 by the American Psychological Association (br). **59 PNAS:** Based on Fig. 2A from "Sex differences in structural connectome," Madhura Ingahlalikar et al., Proceedings of the National Academy of Sciences Jan 2014, 111 (2) 823–828; DOI: 10.1073/pnas.1316909110 (crb). **103 PLoS Biology:** Based on Fig. 4 from "Grasping the Intentions of Others with One's Own Mirror Neuron System," Iacoboni M., Molnar-Szakacs I., Gallese V., Buccino G., Mazziotta J. C., Rizzolatti G., Feb 2005 PLoS Biol 3(3):e79. doi:10.1371/journal.pbio.0030079 (crb). **155 PLoS ONE:** Based on Fig. 3A from "Neural Substrates of Interactive Musical Improvisation: An fMRI Study of 'Trading Fours' in Jazz," Gabriel F. Donnay, Summer K. Rankin, Monica Lopez-Gonzalez, Patpong Jiradejvong, Charles J. Limb, Feb 2014 PLoS ONE 9(2): e88665. https://doi.org/10.1371/journal.pone.0088665 (bc).

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